

AtrialSpectralBench: incremental predictive value and perturbation-theoretic validity of a spectral fragility index for atrial reentry

A pre-registered evaluation on 182 patient-derived left-atrial meshes and 10^5 synthetic excitable networks

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Abstract

We ask whether a closed-form spectral graph biomarker predicts reentry inducibility in atrial tissue beyond established fibrosis and connectivity features. The biomarker is the **Spectral Fragility Index (SFI)**, the Fiedler-value sensitivity $\partial\lambda_2/\partial w_{ij} = (\varphi_i - \varphi_j)^2$ aggregated under a fibrosis-weighted conduction-uncoupling field. Pre-registered before any labels were seen, the primary endpoint was $\Delta\text{AUC} \geq 0.05$ with a DeLong paired $p < 0.05$ on shape-family-held-out folds. *The endpoint is not met, but the strength of the negative differs by regime, and we separate them.* On up to 100,000 synthetic excitable networks SFI adds no practically meaningful predictive value (subspace $\Delta\text{AUC} \approx +0.003$; $\sim 4,150$ cases needed at 80% power, resolvable only because N is huge, and so *clinically undetectable* there). On two real cohorts (182 patients: Roney $n = 100$, UW/Boyle $n = 82$) the combined gradient-boosting estimate is $+0.012$ with a bootstrap interval $[-0.017, +0.044]$ that excludes the threshold, so the real-anatomy result is a measured null rather than a shortfall of power. An earlier version of this work reported $+0.051$ here and called the real-cohort evidence inconclusive. Correcting a defect in one cohort’s released substrate, which we found only afterwards, moved that figure to $+0.036$ at the original cohort size, and extending the Roney arm from 62 to all 100 released meshes moved it to $+0.012$, so most of the drop was sample size rather than substrate. The audit that found the defect, and the four controls that followed, are reported in full; the cause of that cohort’s unusually low inducibility rate turned out not to be any substrate defect but a 30% difference in fibrosis burden between the two released cohorts, which accounts for the apparent anomaly and reproduces it in either direction when the burden is swapped. Where the null is clean, only half the mechanism is feature redundancy: the aggregate SFI summaries are reconstructible from the competitor set at $R^2 = 0.970$, while the extreme and edge-level columns retain variance of their own ($R^2 < 0.25$) and still do not predict the label. The exact per-region $\Delta\lambda_2$ SFI, which has no first-order approximation error, also adds nothing over the competitor set. A separate, label-free result explains why the *single-vector* SFI is also ill-conditioned on real tissue. It is valid only while $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2) = O(1)$, and real atria have median $\rho \approx 2422$, at least $272\times$ past the empirical $\rho^* \approx 3$ on every mesh tested, so by Davis–Kahan the Fiedler vector is unstable. We do *not* claim ρ proves non-predictiveness; it bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a ranking feature. The spectral gap collapses as $N^{-1.05}$ on the real atrial surface (the 2-manifold Weyl exponent), so ρ grows lawfully with resolution. One component partially survives: the Fiedler-gradient localizer $|\nabla\varphi_2|$ is a weak but real rank localizer (origin in the top $\sim 18\%$ on synthetic networks, and weaker on real anatomy, where only ~ 20 origins are available) that beats graph-centrality baselines and transfers across three dynamical media, though it co-localizes with substrate

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imaging rather than beating it, and substrate imaging is the better localizer in the FHN arm (~ 0.10 against 0.176), for a reason we make precise. Naive methodology (random CV, p -value thresholds, no effect-size gate) would have reported SFI as working; our pre-registered protocol rejects it. Our own convergence study surfaces a key limitation: the inducibility labeller is strongly resolution-sensitive and, on a completed six-tier study (1500–48000 nodes, $n = 24$), does not converge within the tested window; the direction of the residual bias at the 2,000-node operating resolution is not established, since the finest tier (37.5%) sits above the operating-resolution rate (34.0%). Our contribution is a carefully bounded negative result: the mechanism behind it, a label-free criterion for the estimator’s validity, a characterization of how it scales, and a demonstration of the false positives that standard methodology produces.

Keywords: algebraic connectivity; Fiedler value; spectral graph theory; atrial fibrillation; reentry; biomarker; perturbation theory; validity radius; pre-registration; negative result.

This is a 100% in-silico computational study. Ground-truth labels are electrophysiology-simulator inducibility verdicts, never clinical outcomes. All numbers trace to `results/*.json` and `notebooks/lab_notebook.md` in the project repository.

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1 Introduction

Algebraic connectivity λ_2 (Fiedler 1973 [1]) governs the stability of coupled excitable and oscillator networks (the master-stability-function result [3] and the synchronizability eigenratio λ_N/λ_2 [4]), and has been used as a real-time seizure detector [5]. The per-edge sensitivity

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2 \quad (1)$$

is a classical identity underlying Fiedler-vector edge-editing heuristics [2]. It is therefore natural to hypothesize that aggregating this sensitivity over a fibrosis-weighted perioperative uncoupling field yields a per-region biomarker of atrial-fibrillation (AF) reentry fragility. We test this hypothesis and *pre-register the null* (“SFI merely re-encodes fibrosis”) as an accepted, reportable outcome, so that a negative result is a success of the protocol rather than a file-drawer casualty.

The mathematics we use is classical; we claim no new theorem. Our contributions are (i) a leakage-controlled incremental test of SFI against real competitors and a spatial null on real anatomy; (ii) a label-free validity criterion and a measured gap-scaling exponent that matches the Weyl 2-manifold prediction on real atria, which together explain why the *single-vector estimator* is ill-conditioned on real tissue and why any first-order Fiedler-based marker faces the same conditioning problem there, and which do not explain the predictive null; (iii) a cross-medium transfer analysis isolating the one component that generalizes; and (iv) a falsification-protocol demonstration showing that standard methodology would have manufactured exactly the false positive this protocol catches.

In brief, a closed-form spectral fragility index does *not* add practically meaningful reentry-inducibility prediction beyond existing features, either at synthetic scale or on 182 patients of real anatomy. The mechanism is partly feature redundancy and partly a feature that is genuinely new but unpredictable (Section 4.2). Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). What we report is not a discovery but a carefully drawn boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how readily naive methodology mistakes redundant, inconclusive signal for a working biomarker.

2 Theory: the Spectral Fragility Index

The core mathematics is classical numerical linear algebra; the contribution is its *use*. We state each result with its prior art (Section 2.5) and make the validity boundary explicit (Section 2.4).

2.1 The weighted conduction graph

Nodes are mesh vertices; edges are triangle edges; the edge weight $w_{ij} > 0$ is a fibre-anisotropic conduction conductance attenuated by local fibrosis. The combinatorial Laplacian is

$$L = D - W, \quad D = \text{diag} \left(\sum_j w_{ij} \right). \quad (2)$$

L is symmetric positive-semidefinite with a single zero eigenvalue (the constant vector) for a connected graph, with spectrum $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$ and eigenvectors $\varphi_1, \varphi_2, \dots$; λ_2 is the algebraic connectivity and φ_2 the Fiedler vector.

2.2 The closed-form fragility (first order)

L is a sum of rank-one edge terms, so

$$L = \sum_{(i,j) \in E} w_{ij} (e_i - e_j)(e_i - e_j)^\top \implies \frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^\top. \quad (3)$$

For a *simple* eigenvalue λ with unit eigenvector φ , first-order (Hellmann–Feynman) perturbation theory gives $\partial\lambda/\partial w_{ij} = \varphi^\top (\partial L/\partial w_{ij}) \varphi = (\varphi_i - \varphi_j)^2$, which for the Fiedler pair is Eq. (1): a non-negative per-edge *fragility*. Reducing a conductance w_{ij} by a small $\Delta w_{ij} \geq 0$ (uncoupling) lowers λ_2 by $\Delta w_{ij}(\varphi_{2,i} - \varphi_{2,j})^2$ to first order.

The identity is verified in six independent ways rather than merely asserted. Computer algebra: (i) sympy proves Eqs. (1)–(3) symbolically, and (ii) on a concrete 5-node graph the finite-difference $\partial\lambda_2/\partial w_{ij}$ matches $(\varphi_{2,i} - \varphi_{2,j})^2$ to $|\text{err}| \approx 2 \times 10^{-17}$ (mpmath, 40 digits). Exactness: (iii) the identity holds *exactly* with rational weights (zero floating point); (iv) an interval-arithmetic enclosure certifies the true λ_2 to within $\sim 4 \times 10^{-40}$; and (v) λ_2/φ_2 agree across scipy, networkx, and a hand-rolled inverse iteration (spread 2.5×10^{-15}). These three tolerances are printed at run time by `scripts/verify_sfi_identity.py` and `scripts/verify_math_rigor.py` rather than stored in `results/`, so they are reproduced by rerunning those scripts. Finally, (vi) the algebraic core is kernel-checked in the Coq proof assistant over the exact rationals, axiom-free (Appendix A). The analytic Hellmann–Feynman step (differentiability of a simple eigenvalue) is the cited classical result [2], not re-proved here.

2.3 Per-region aggregation: the SFI

Model diffuse perioperative stress as a distribution of small, non-negative, fibrosis-weighted fractional conductance reductions Δw (mean $\mathbb{E}[\Delta w_{ij}] = \text{frac}_{ij} w_{ij}$, calibrated to literature perioperative conduction-velocity slowing; $\Delta w \approx 0.36$, frozen in the pre-registration). The per-region Spectral Fragility Index is the expected λ_2 drop contributed by region R :

$$\text{SFI}(R) = \mathbb{E}[\Delta\lambda_2 \text{ from } R] \approx \underbrace{\sum_{(i,j) \in R} \mathbb{E}[\Delta w_{ij}] (\varphi_{2,i} - \varphi_{2,j})^2}_{\text{first order}} + \underbrace{\sum_{k \neq 2} \frac{(\varphi_k^\top \Delta L \varphi_2)^2}{\lambda_2 - \lambda_k}}_{\text{2nd-order resolvent}} + O(\|\Delta L\|^3). \quad (4)$$

The second-order term is the source of the validity radius below. Near eigenvalue degeneracy ($\lambda_2 \approx \lambda_3$) the single φ_2 is an arbitrary rotation in the low invariant subspace, so we use a basis-independent *subspace (projector) SFI*: with $P = \sum_c \varphi_c \varphi_c^\top$ over the clustered subspace, $s_{ij}^\top P s_{ij} = \sum_c (\varphi_{c,i} - \varphi_{c,j})^2$ ($s_{ij} = e_i - e_j$) is well-defined. For discrete edits (Maze lesion, atriotomy) the expansion is abandoned and λ_2 is recomputed *exactly* after the rank-one edit.

2.4 The validity radius (the central caveat)

The first-order form is trustworthy only while the perturbation is small *relative to the spectral gap* $\lambda_3 - \lambda_2$. Two different tools bound two different things (a common error is to use Weyl for both):

- Weyl’s inequality bounds the *eigenvalue* move: $|\Delta\lambda_2| \leq \|\Delta L\|$.
- Davis–Kahan [7] bounds the *eigenvector* move, which scales like $\|\Delta L\|/(\lambda_3 - \lambda_2)$. This, not Weyl, governs when φ_2 (hence the single-vector SFI) degrades.

Define the *validity radius*

$$\rho := \frac{\|\Delta L\|}{\lambda_3 - \lambda_2}. \quad (5)$$

From second-order Rayleigh–Schrödinger theory the relative error of the first-order $\Delta\lambda_2$ prediction is $\Theta(\rho)$, and the Fiedler vector rotates by $\sin \angle(\varphi_2, \varphi'_2) \leq \sqrt{2} \rho$ (Davis–Kahan). Empirically

the error is 1% at $\rho = 0.3$, crosses 10% at $\rho^* \approx 3$, and reaches 92% at $\rho = 28$, with Weyl holding throughout. On real excitable media the gap is tiny, so ρ is enormous. Over the 12 scanned meshes the median is $\rho \approx 2422$ with an interquartile range of [1082, 7384] and a full range of [817, 16554]; the spread is wide because ρ inherits the near-degenerate gap in its denominator, but every mesh exceeds ρ^* and even the smallest does so by a factor of 272, so the conclusion does not rest on the median. That is about $10^3 \times$ past ρ^* .

The criterion ρ proves that the *single-vector* first-order SFI is a numerically *invalid estimator* of $\Delta\lambda_2$ on real tissue, and that φ_2 is ill-conditioned (Davis–Kahan), a label-free statement about the estimator. It does *not* prove SFI is non-predictive: ρ bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a monotone ranking feature; indeed the *exact* $\Delta\lambda_2$ SFI (no ρ limitation) is *also* non-predictive (Section 4.1), so the predictive null has a separate cause, which Section 4.2 measures and finds to be only half redundancy. Two caveats: (i) ρ ’s numerator carries the fibrosis-weighted Δw field, so only its denominator $\lambda_3 - \lambda_2$ is truly label-free; (ii) $\rho^* \approx 3$ is calibrated on a 16-node path graph with a problem-dependent constant, so “ $10^3 \times$ past ρ^* ” is order-of-magnitude, not exact.

2.5 Prior art (this is not new mathematics)

The identity (1) is Ghosh & Boyd (2006) [2]; algebraic connectivity is Fiedler (1973) [1]; λ_2 governing coupled excitable/oscillator stability is Pecora & Carroll (1998) [3] with the eigenratio of Barahona & Pecora (2002) [4]; Weyl’s law, Davis–Kahan, and second-order perturbation theory are classical (1970s and earlier). No result in Section 2 is a new theorem, and none is presented as one. The contribution is recognizing that ρ is the right *a priori* validity screen for a spectral fragility biomarker, measuring its violation on real tissue, and using it to *mechanistically explain* a negative result.

3 Methods

Full detail is in the pre-registration (docs/PRE_REGISTRATION.md) and theory note (docs/SFI_THEORY.md); the endpoints, cross-validation scheme, baseline set, and accepted null were frozen before any labels were inspected.

The edge weight. Everything downstream — λ_2 , the Fiedler vector, the spectral gap, ρ and the SFI itself — is a function of the conduction graph’s edge weights, so we give their form explicitly. For an edge (i, j) let $\hat{e}_{ij}$ be the unit vector along it, $\bar{f}_{ij}$ the normalised mean of the two endpoint fibre directions, and $a_{ij} = |\hat{e}_{ij} \cdot \bar{f}_{ij}|$ their absolute alignment. With $\phi_{ij} = \frac{1}{2}(\phi_i + \phi_j)$ the mean endpoint fibrosis,

$$w_{ij} = \underbrace{[c_{\perp} + (c_{\parallel} - c_{\perp}) a_{ij}^2]}_{\text{anisotropy}} \cdot \underbrace{\text{clip}(1 - \phi_{ij}, \varepsilon, 1)}_{\text{fibrosis}}, \quad (6)$$

with $c_{\parallel} = 1.0$ along fibre, $c_{\perp} = 0.3$ across, and a floor $\varepsilon = 0.05$ so dense scar retains a small residual conductance rather than disconnecting the graph. The a^2 dependence makes the weight a function of alignment rather than of direction, so it is invariant to fibre sign, which is the only property of $\bar{f}$ that is identifiable. Note that both factors are bounded, so $w_{ij} \in [\varepsilon c_{\perp}, c_{\parallel}]$; no edge weight can vanish and the graph stays connected by construction.

Cohort flow. The Roney deposit contains 100 meshes and all 100 are used. The UW deposit contains 164 meshes, which are 82 patients imaged twice, before and after ablation; the primary analysis uses the 82 *pre-ablation* meshes only, since post-ablation geometry encodes the delivered treatment. No subject is excluded for any other reason, so the analysed cohort is $100 + 82 = 182$

and there is no attrition to report. The 164-mesh figure appears in this paper only where a statement concerns the released files themselves, such as the fibre-field and element-tag audits.

Inducibility protocol. Labels come from a monodomain Mitchell–Schaeffer solve on the coarsened surface (2000 nodes, vertex clustering). Membrane parameters, frozen before any label was generated: $\tau_{\text{in}} = 0.3$, $\tau_{\text{out}} = 6.0$, $\tau_{\text{open}} = 120$, $\tau_{\text{close}} = 110$ ms and $v_{\text{gate}} = 0.13$, with fibrosis shortening τ_{close} by up to 50%. Diffusion uses an anisotropic conductance-weighted cotangent Laplacian with $d_0 = 0.20$ and $w_{\text{ref}} = 0.3$, integrated by backward Euler at $\Delta t = 0.05$ ms. Induction is burst pacing from 2 randomly chosen sites, 6 beats at a cycle length of 150 ms, stimulus amplitude 0.15 over 2 ms within a 4 mm radius. A subject is labelled inducible if self-sustained supra-threshold activity persists at least 650 ms after the last stimulus, observed over a 1000 ms window. The protocol was verified before use: APD90 218 ms healthy and 121 ms fibrotic, planar conduction velocity 0.4–1.2 m/s, and a monotone relationship between fibrosis and sustained reentry.

Evaluation pipeline. Cross-validation is `GroupKFold` with the group set to the patient, so no subject appears in both training and test folds, using $k = \min(5, n_{\text{groups}})$ folds. Two classifiers are run. Logistic regression is a standardisation-plus-`LogisticRegression` pipeline (`liblinear`, at most 2000 iterations) whose regularisation C is chosen in an *inner* `GroupKFold` over $\{0.03, 0.1, 0.3, 1.0, 3.0\}$, so tuning never sees the outer test fold. Gradient boosting is deliberately *not* tuned — 120 trees, depth 2, learning rate 0.05, subsample 0.8 — because at these sample sizes a search would fit the folds; this is a fixed, shallow reference rather than a competitive model. We log it as a deviation from the pre-registered nested selection, which specifies an inner `GroupKFold` search for both classifiers. Out-of-fold predicted probabilities are pooled across folds and compared by the DeLong test. Pooling out-of-fold scores means the two compared vectors come from k different fitted models rather than from two fixed scoring functions, which is DeLong’s assumption; the paired structure and the leakage-free grouping are preserved, but the resulting p -values should be read as approximate. The naive comparison reported alongside replaces the grouping with shuffled `KFold`. On the real cohorts each patient contributes one row and is its own group, so `GroupKFold` there coincides with unshuffled `KFold` and the grouping is a guarantee rather than a correction; the leakage it exists to prevent is demonstrated on the replicated synthetic cohort, where resamples of a single shape family have several members (Section 4.10).

Substrate. Real left-atrial surfaces are loaded into weighted conduction graphs from two public cohorts: the Roney LA virtual cohort (Zenodo 5801337 [11]) and the UW/Boyle LGE-MRI cohort (Dryad 10.5061/dryad.kkwh70sg0, 82 distinct AF patients [12]). Each mesh carries an image-intensity-ratio (IIR) fibrosis proxy mapped to a fibrosis field by the Khurram healthy-to-scar ramp [13]. The two cohorts differ in what else they carry: the Roney meshes ship registered universal atrial coordinates and per-vertex fibre fields (directional spread 0.945–0.979), whereas the UW meshes ship a `VECTORS fiber` array that is the constant $(1, 0, 0)$ everywhere (spread 0.000) and no atrial coordinates, for which we substitute a PCA chart, so UW anisotropy is a fixed coordinate bias rather than anatomy (Sections 4.3 and 4.4). Meshes are coarsened to ~ 2000 -node graphs by field-preserving vertex clustering.

Labels. Ground truth is a monodomain Mitchell–Schaeffer reaction–diffusion inducibility verdict [10] (self-sustained reentry ≥ 650 ms after burst pacing at cycle length 150 ms from random sites), run on each real anatomy with an anisotropic conductance-weighted cotangent Laplacian and implicit time stepping. Reentry here is governed by wavelength ($\text{CV} \times \text{ERP}$), source–sink mismatch, and unidirectional block, *never* by λ_2 , so a predictive $\text{SFI} \rightarrow \text{label}$ link is not circular. These are monodomain simulator verdicts, not clinical post-operative AF. openCARP is

run alongside on all 182 subjects as a robustness check rather than as the labeller (`results/opencarp_full_cohort.json`), and agrees with the monodomain verdict on 156/182.

The SFI feature and its competitors. Per-region SFI is computed as in Eq. (4) (first-order, subspace, and exact-recompute variants). It is evaluated strictly as an add-on/ablation against a non-trivial competitor set that it must beat: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and λ_2 -alone (isolating what the *SFI construction* adds over the bare algebraic connectivity).

Evaluation protocol (the guards). Pre-registered: shape-family GroupKFold (each patient is its own group, so cross-validation is patient-held-out; grouped *and* naive AUC both reported so leakage is visible), the DeLong paired test [8] in its fast midrank form [9], $\geq 10^4$ stratified bootstrap resamples (a logged deviation: the Kuramoto transfer arm was run at 5×10^3), a rotational/torus-shift spatial null for localization, tie-robust permutation nulls, and a keep/delete rule deciding each sub-claim by *measured* correlation. Effect sizes are gated, not just p -values. The primary endpoint requires *both* grouped $\Delta\text{AUC} \geq 0.05$ *and* DeLong $p < 0.05$.

Synthetic media, generative model. The synthetic cohorts are random geometric graphs, which give a two-dimensional excitable sheet with the degree heterogeneity of real tissue and none of its anatomy. Each network draws n points uniformly on the unit square and connects every pair within a fixed radius; if that radius leaves the graph too sparse the construction falls back to a symmetric 6-nearest-neighbour graph, and in either case only the largest connected component is retained, so the medium is a single sheet rather than several. Lesion burden is swept uniformly over $[0.05, 0.45]$ across the cohort, which is what makes both classes appear: at the low end the network is stable and at the high end it is not, mirroring the mild-to-severe fibrosis gradient of the atrial cohorts. Because the topology is generated rather than measured, these cohorts carry no fibre field and no anatomical coordinates, and they are used only where those are irrelevant — for the scale behaviour of the null and for the cross-medium transfer of the localizer.

Transfer media. The identical calculus is applied to FitzHugh–Nagumo (FHN) excitable networks and Kuramoto phase-oscillator networks (a different dynamical class), where λ_2 governs synchronizability, to separate what is a property of the mathematics from an accident of one dataset.

4 Results

4.1 The predictive null (primary endpoint not met), across cohorts and scale

Table 1: Incremental predictive value of SFI over the full competitor set (`results/gm1_expanded_metrics.json`; the cohort-size-matched comparison is `results/gm1_matched_n.json`). ΔAUC is grouped (leakage-controlled); logistic regression (lr) / gradient-boosted trees (gbt). The endpoint (grouped $\Delta\text{AUC} \geq 0.05$ and DeLong $p < 0.05$) is met in no regime, and every interval on real anatomy contains zero. The UW rows rest on eight positives and are uninformative rather than null; note in particular that their base AUCs differ sharply between the competitor and fibrosis-only baselines, which is what inflates the latter’s apparent increment.

cohort / medium	events/ N	model	base AUC	SFI ΔAUC (95% CI); p
Roney (real)	34/100	lr	0.870	-0.021 $[-0.075, +0.033]$; 0.44
		gbt	0.856	$+0.009$ $[-0.029, +0.046]$; 0.63
UW/Boyle (real)	8/82	lr	0.834	$+0.000$ $[-0.073, +0.081]$; 1.00
		gbt	0.748	$+0.035$ $[-0.095, +0.203]$; 0.67
Combined real	42/182	lr	0.884	-0.007 $[-0.042, +0.029]$; 0.70
		gbt	0.851	$+0.012$ $[-0.017, +0.044]$; 0.45
FHN networks	2,000 networks			$\leq +0.003$; $p \geq 0.12$
FHN networks	up to 100,000			subspace $+0.0028$; $p < 10^{-4}$ at $N \geq 10^4$
Kuramoto networks	500 networks			$\leq +0.0036$; $p \geq 0.38$

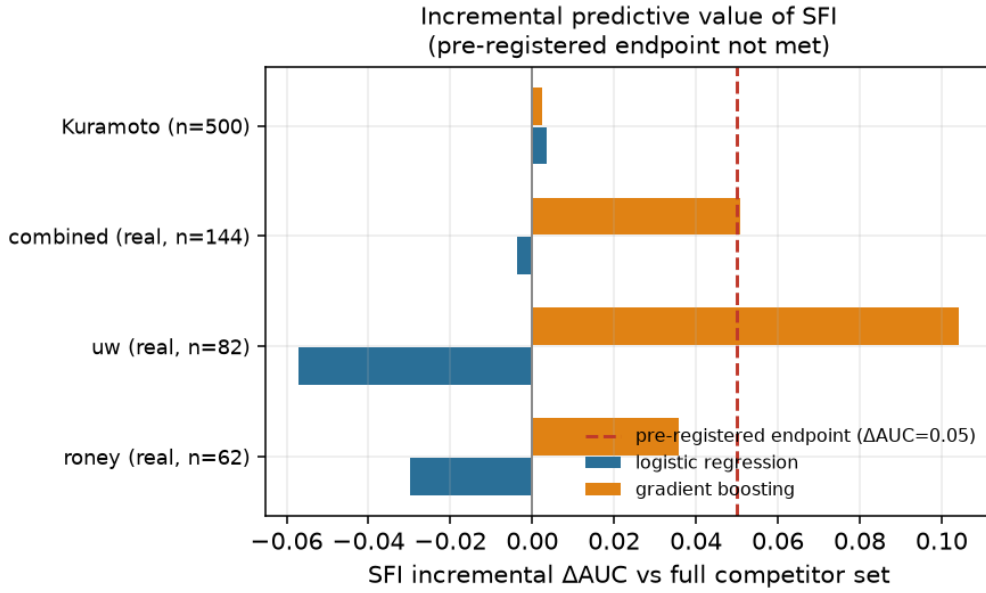


Figure 1: The predictive null. Adding SFI to the competitor set does not clear the pre-registered effect-size gate in any cohort. On Roney and on the combined cohort the bootstrap interval excludes the gate as well as containing zero, so those are measured nulls rather than a shortfall of power; the two UW intervals reach past the gate ($[-0.073, +0.081]$ and $[-0.095, +0.203]$ on eight positives) and are uninformative rather than null.

The negative is not uniform, and we separate the regimes. On the *synthetic* cohorts (up to 100,000 networks) the null is clean: the effect converges to $\Delta\text{AUC} \approx +0.003$, statistically resolvable only because N is enormous. On the *real* cohorts it is a measured null. The combined

gradient-boosting point estimate is $+0.012$ (DeLong $p = 0.449$, bootstrap CI $[-0.017, +0.044]$), roughly a quarter of the pre-registered 0.05 threshold and with an interval that excludes it. An earlier version of this analysis reported $+0.051$ here and had to argue that the endpoint was missed only on significance. That estimate was an artefact of the sealed UW substrate: correcting it drops the figure to $+0.036$ at the original cohort size and to $+0.012$ once the Roney arm is extended to all 100 released meshes. With 42 inducible of 182 this is the largest real-anatomy arm in the study, and the null here is a measurement rather than a shortfall. The combined row is a single pooled ROC across two cohorts of unequal prevalence (34/100 on Roney, 8/82 on UW), so its ranking carries between-cohort as well as within-cohort separability; that is why the pooled base AUC under logistic regression, 0.884, exceeds both components (0.870 and 0.834), and the pooled increment should be read alongside the cohort-specific estimates rather than in place of them.

The UW cohort remains the weak arm, though not for the reason we previously gave. On eight positives its intervals are wide, the competitor comparison spanning $[-0.095, +0.203]$ under gradient boosting, so it is an underpowered external check of the *labeller* rather than of SFI. We had also described its competitor baseline as close to chance, and that was simply wrong: the measured competitor base AUC is 0.834 under logistic regression and 0.748 under gradient boosting. Only the *fibrosis-only* baseline approaches chance, at 0.562, and it is that near-chance baseline which manufactures the cohort's largest apparent effect, $\Delta\text{AUC} = +0.217$ with $p = 0.121$ and an interval $[-0.030, +0.485]$ that spans half the available range. A large ratio over a baseline that predicts almost nothing, on eight events, is not a signal.

The null is not explained by the validity radius. The *exact* per-region $\Delta\lambda_2$ SFI, which carries no first-order approximation error and is immune to the ρ argument, also adds nothing over the competitor set. Our reading is that SFI's information is already present in standard connectivity features (λ_2 -alone, min-cut, percolation), but an incremental null does not by itself separate that redundancy from a feature that carries little predictive information at all. We separate the two in Section 4.2, directly and without reference to the label, where the redundancy account holds for the region-level aggregates and fails for the extreme and edge-level columns. Either way the incremental null is itself a label-dependent finding, separate from ρ .

Fibrosis-heterogeneity baseline comparison (pre-specified as an additional feature-set contrast), and its withdrawal. This is not the pre-registered secondary endpoint, which is the localization test reported in Section 4.9; it was pre-specified as a further feature-set comparison alongside the primary one. An earlier version of this analysis reported that SFI beat a *fibrosis-heterogeneity-only* baseline on the combined real cohort (GBT $\Delta\text{AUC} = +0.103$, $p = 0.009$) and read that as evidence SFI is not merely re-encoded fibrosis but carries connectivity information beyond the substrate. That result does not survive, and we withdraw both it and the claim it supported. Two changes separate the old number from the new one, and they do not contribute equally; both figures below are gradient-boosting ΔAUC against the same fibrosis-heterogeneity baseline, tested with the paired DeLong statistic. Correcting the UW substrate while holding the cohort at its original 144 subjects leaves the estimate largely intact ($+0.093$, $p = 0.013$), so this was not principally a substrate artefact. Extending the Roney arm from 62 to the full 100 released meshes is what collapses it, to $\Delta\text{AUC} = +0.017$ ($p = 0.452$). An effect that disappears when 38 further real subjects are added was a small-sample fluctuation, and it was one of ~ 30 DeLong comparisons besides. On the full cohort no comparison, primary or secondary, shows SFI adding information beyond either baseline.

The discarded estimate is worth one more sentence, because its failure mode is instructive. Under the sealed substrate the UW arm alone returned $\Delta\text{AUC} = +0.340$ ($p = 0.026$) on six positives, a value no one should have believed and which we did not flag at the time; it is that arm which carried the combined figure. A large point estimate resting on six events is a warning, not a finding.

Naive (ungrouped) comparison. The pre-registration asks for the naive Δ AUC beside the grouped one so that leakage is visible, and we give it for every real-cohort arm: Roney +0.017 (lr) and -0.011 (gbt), UW +0.066 and -0.160, combined -0.011 and +0.010, against the grouped values of Table 1. The gap has no consistent sign: the naive estimate runs above the grouped one in two of the six arms, both logistic regression on a single cohort, and below it in the other four including the pooled logistic arm, so nothing here reproduces the optimism ungrouped cross-validation is expected to show. On these cohorts each patient contributes one row and is its own group, so the grouping is close to a no-op. The leakage effect is demonstrated instead on the replicated synthetic cohort, where resampled networks share a parent shape family.

Power analysis. In the synthetic cohort the subspace effect (+0.0028) needs $\sim 4,150$ cases for 80% power and the single-vector effect (+0.0007) $\sim 23,000$, clinically undetectable *there*. We do *not* transport “clinically undetectable” to the real cohorts, which are a separate regime. On the corrected substrate their estimates are +0.009 (Roney, $n = 100$) and +0.012 (combined, $n = 182$), with intervals that contain zero and exclude the 0.05 gate. An earlier version read the then-current +0.036–0.051 as compatible with a detectable effect we were underpowered to confirm; those estimates are superseded and that reading no longer holds.

4.2 Testing the redundancy explanation directly

Source: `results/feature_redundancy.json`.

The account above attributes the null to feature redundancy. That explanation was inferred from the incremental AUC being zero, which does not establish it: a feature that is pure noise also adds nothing, and so does one the classifier cannot exploit. Those are three different diagnoses. We therefore measured redundancy directly and without reference to the label, by regressing each SFI column on the competitor block and by taking the leading canonical correlation between the two blocks, with a permuted-SFI control to separate genuine redundancy from a degenerate estimator.

The blocks are almost collinear in their dominant direction: the leading canonical correlation between the nine SFI columns and the six competitors is 0.996, against 0.289 for the permuted control. Taken alone that supports the redundancy account. Per column it is less uniform, and the structure matters:

SFI column	R^2 on competitors	R^2 permuted
<code>sfi_total, sfi_region_mean</code>	0.970	0.008
<code>sfi_top3</code>	0.842	0.022
<code>sfi_top2</code>	0.665	0.040
<code>sfi_edge_frag_max</code>	0.205	0.010
<code>sfi_edge_frag_mean</code>	0.154	0.015
<code>sfi_region_max, sfi_top1</code>	0.138	0.007
<code>sfi_region_std</code>	0.098	0.006

Four of the nine columns are reconstructible from the competitors at $R^2 \geq 0.60$, and the two aggregate summaries almost perfectly so. The remaining five keep most of their variance, at $R^2 < 0.25$, well above the permuted control but far from redundant. The split does not fall along the aggregate versus order-statistic line: what the competitors reconstruct is the two region-level aggregates and the deeper region order statistics `sfi_top2` and `sfi_top3`, and what resists reconstruction is the edge-level pair, the region maximum and the region spread.

This qualifies rather than overturns the null, and it sharpens what the null means. For `sfi_total` and `sfi_region_mean` the competitors already carry the information, and adding them cannot help. For `sfi_region_max`, `sfi_top1` and `sfi_region_std` there is genuine unique variance in the feature, and the finding is the stronger one: that variance exists and does not

predict the label. A biomarker can fail either by saying nothing new or by saying something new that does not matter, and on this evidence SFI does both, in different components. We had previously reported only the first.

4.3 Region definition and the patient-level reduction

The per-region SFI of Eq. (4) is stated over a region R , and until now we had left R implicit. Since the partition and the reduction that follows it are what actually produce the feature block whose incremental ΔAUC is the primary endpoint, we give both explicitly. Neither was pre-registered — the pre-registration refers to the per-region SFI throughout without fixing a partition or a patient-level reduction — so both are a logged deviation, fixed post hoc but without reference to the outcome, since SFI is closed-form given the mesh and is never fitted to the label.

One further deviation belongs here because it bears directly on the validity argument. Every real-cohort number in this paper is computed from the *first-order* per-region SFI (`asb.sfi.sfi_region`). The pre-registration also specifies a gap-triggered switch to the subspace variant when the validity radius is exceeded, and that switch is not wired into the real-cohort feature path: the subspace estimator is used only in the synthetic validity experiments of Section 2.4. Since real atria sit at median $\rho \approx 2422$, the switch would have fired on every mesh had it been applied. We report the first-order numbers rather than silently re-deriving the table, for two reasons: the ranking a classifier consumes is invariant to the global Δw magnitude that drives ρ , and the question the switch exists to answer is settled directly in Section 4.1, where the *exact* per-region $\Delta\lambda_2$ — which carries no first-order error at any $\|\Delta L\|$ — also adds nothing over the competitor set. The reader should nonetheless know that the primary endpoint was scored with the estimator the validity argument indicts.

Partition. Regions are a 3×3 grid over the universal atrial coordinates. With $(\alpha, \beta) \in [0, 1]^2$ per vertex, a vertex is assigned $r = 3\lfloor 3\beta \rfloor + \lfloor 3\alpha \rfloor \in \{0, \dots, 8\}$, so every mesh carries exactly nine regions regardless of its vertex count. An *edge* is assigned region r only when both endpoints carry r ; edges straddling two regions go to a tenth bucket, $r = -1$, which is retained rather than discarded so that the per-edge sensitivities sum to the global total. The rule is shared by the closed-form and Monte-Carlo estimators so their region keys coincide.

This is where the two cohorts differ in kind and not only in degree. The Roney meshes carry anatomically registered universal atrial coordinates, so their nine regions are anatomical. The UW release ships none, so we substitute the top two principal axes of the surface coordinates, min-max normalised. That surrogate is a smooth two-dimensional chart adequate for binning, but it is not registered to anatomy, and any region-level or localization claim on the UW cohort is correspondingly weaker. The substitution is recorded in the mesh metadata rather than left to the reader to infer.

Reduction to the patient. The classification target is one binary verdict per patient, so the nine per-region values must be reduced. Nine scalars enter the design matrix, all in the `sfi_` namespace: the global `total` (the sum of expected per-edge λ_2 drop over all edges); `region_max`, `region_mean` and `region_std` over the region values; the three largest region values in descending order, `top1`–`top3`; and `edge_frag_max` and `edge_frag_mean` over the raw per-edge fragilities $(\varphi_{2,i} - \varphi_{2,j})^2$. The order statistics are included because a fragility biomarker is a claim about the *worst* region rather than the average one, and reporting only a mean would have made the hypothesis untestable in its own terms. Two of the nine names denote one quantity: `region_max` and `top1` are both the maximum over the region values, and are identical on all 182 subjects to zero difference, so the block carries eight distinct columns under nine names. That is why they share a row of the table in Section 4.2.

Two properties of this construction matter for reading the null. First, the competitor features are patient-level scalars already, so no analogous reduction is applied to them; the comparison is between an add-on of eight distinct columns and a six-column baseline, which if anything favours the add-on. Second, none of the nine columns is fitted to the outcome. SFI is closed-form given the mesh and the frozen uncoupling field, so it contributes no tuning degrees of freedom, and the null cannot be attributed to a failure of feature selection.

4.4 A provenance audit of the released substrates, and four controls

Sources, all under `results/`: the cohort audit is `cohort_geometry_audit.json`, and the four controls are `uw_substrate_ablation.json`, `roney_fibre_control.json`, `roney_fibrosis_quantization.json` and `fibrosis_burden_swap.json`. Every Roney inducibility arm in this section runs on the 62-mesh subset that predates the extension of the predictive cohort to all 100 released meshes, and was not re-run on the full deposit; the predictive results of Section 4.1 use all 100.

The UW cohort forced this section. Under a protocol frozen before any label was generated, and identical in every parameter, the Roney meshes returned 20/62 subjects inducible (32.3%) while the UW meshes returned 6/82 (7.3%) on the substrate as released. Three UW rates recur below and are kept distinct throughout: 6/82 as released, 8/82 after the cap repair — the substrate every predictive result uses — and 14/82 after both repairs. A fourfold discrepancy on cohorts we had assumed carried comparable fibrosis burden — an assumption Control 4 below overturns — is not a cohort property one should accept; it is a hypothesis about the pipeline. We had previously recorded the UW arm as uninformative and moved on, which was the error. What follows is what a proper audit found, then three controls that do not explain the anomaly, and a fourth that does.

The audit. Two defects sit in the released files. The first is ours. The only per-element scalar the UW release carries is `elemTag`, the Dryad README does not define its values, and we had been mapping tag 164 to half-fibrotic tissue. Four measurements say it is not tissue at all. It resolves into four to six large connected components in every one of the 82 meshes, five in 75 of them, four in five and six in two, which is the ordinary variation of pulmonary-vein anatomy: a left common trunk gives four, a right middle vein gives six. Each large component is topologically a disc, 11–42 mm across, sitting at 0.47–0.82 of the atrial radius from the centroid. It borders fibrotic elements on about 0.08% of its incident edges against a chance rate of 14–20%, a depletion beyond a hundredfold, which is incompatible with reading it as a fibrosis border zone. And it is ablation-invariant, holding at 20.0% of elements before the procedure and 19.9% after, while every other class moves. Tag 164 is the set of caps sealing the pulmonary veins and the mitral valve. Removing it opens the surface: the Euler characteristic falls by a median of 7 (range 2–12). We report the *change* rather than an opening count, because $\chi = 2 - k$ presumes a clean closed surface and two of these meshes, ID040 and ID049, arrive at $\chi = -16$ and -18 before anything is removed.

Treating those caps as half-conducting tissue does more than inflate fibrosis. It seals the atrium’s orifices, so simulated activation crosses the mitral valve and the vein ostia instead of circling them, and reentry anchored on those orifices, a principal mechanism of atrial fibrillation, cannot form.

The second defect is upstream. A `VECTORS fiber` array is present in all 164 released meshes, but it equals the constant (1, 0, 0): the measured mean directional spread is exactly 0.000. The Roney cohort, by contrast, gives 0.945–0.979, with roughly one distinct direction per vertex. Conduction anisotropy is applied relative to the *local* fibre direction, so a single global direction reduces it to a fixed coordinate bias carrying no anatomy. The same cardinality argument applies to fibrosis: three distinct levels in the UW release, effectively two once the caps are dropped, against 276–1071 in Roney.

Control 1: repairing the substrate. Opening the orifices and supplying a spatially varying fibre field, in a 2×2 design over all 82 subjects (Fig. 2a), raises inducibility from 6/82 to 14/82. Either repair alone converts two subjects ($6 \rightarrow 8$ in both single-factor arms) and together they convert eight, so on these four cells the averaged main effect of a repair and the interaction contrast are both +4 subjects and cannot be told apart; we do not claim the two act super-additively. The mechanistic reading an interaction would invite — that reentry needs both an anatomical obstacle and anisotropic heterogeneity — is in any case undercut by Control 2 below, which shows the fibre factor does not move the rate at all on a cohort that has a real fibre field. We do not present this as a fix. The strongest contrast does not reach paired significance (McNemar exact $p = 0.077$), it closes only 39% of the distance to Roney, and only two of the original six positives survive the repair. The varying fibre field is moreover a probe, not a correction: it is the global $+z$ direction projected onto each vertex tangent plane, and it is not anatomy.

Control 2: destroying the fibres on a cohort that has them. The cleaner experiment replaces the Roney fibre field with the same constant $(1, 0, 0)$, holding anatomy, fibrosis and every solver parameter fixed (Fig. 2b). The rate does not move at all: 20/62 with real fibres and 20/62 without, McNemar $b = c = 5$, $p = 1.0$. Fibre degeneracy does not drive the inducibility rate, and our first attribution of the UW anomaly to it was wrong. The perturbation is real at the level of the dynamics — sustained reentry duration changes on 50/62 subjects, by up to 661 ms, and 10/62 verdicts flip — but those effects cancel in aggregate.

Control 3: destroying the fibrosis gradation. The remaining structural difference was that UW fibrosis is binary where Roney’s is graded, and graded border zones are where slow conduction and unidirectional block are supposed to arise. Binarising the Roney fibrosis field should therefore have suppressed reentry. It does the opposite (Fig. 2c). Thresholding at $f > 0.5$ raises inducibility from 20/62 to 31/62; thresholding at the per-subject quantile that holds mean fibrosis burden fixed at 0.333, so that only the gradation is destroyed, raises it to 40/62. Coarsening the fibrosis field roughly doubles inducibility. In hindsight the mechanism is unmysterious — a sharp binary boundary is a stronger source–sink discontinuity than a smooth ramp — but the direction is the opposite of our prediction, and it disqualifies coarse fibrosis as an explanation for a cohort that is inducible *too rarely*.

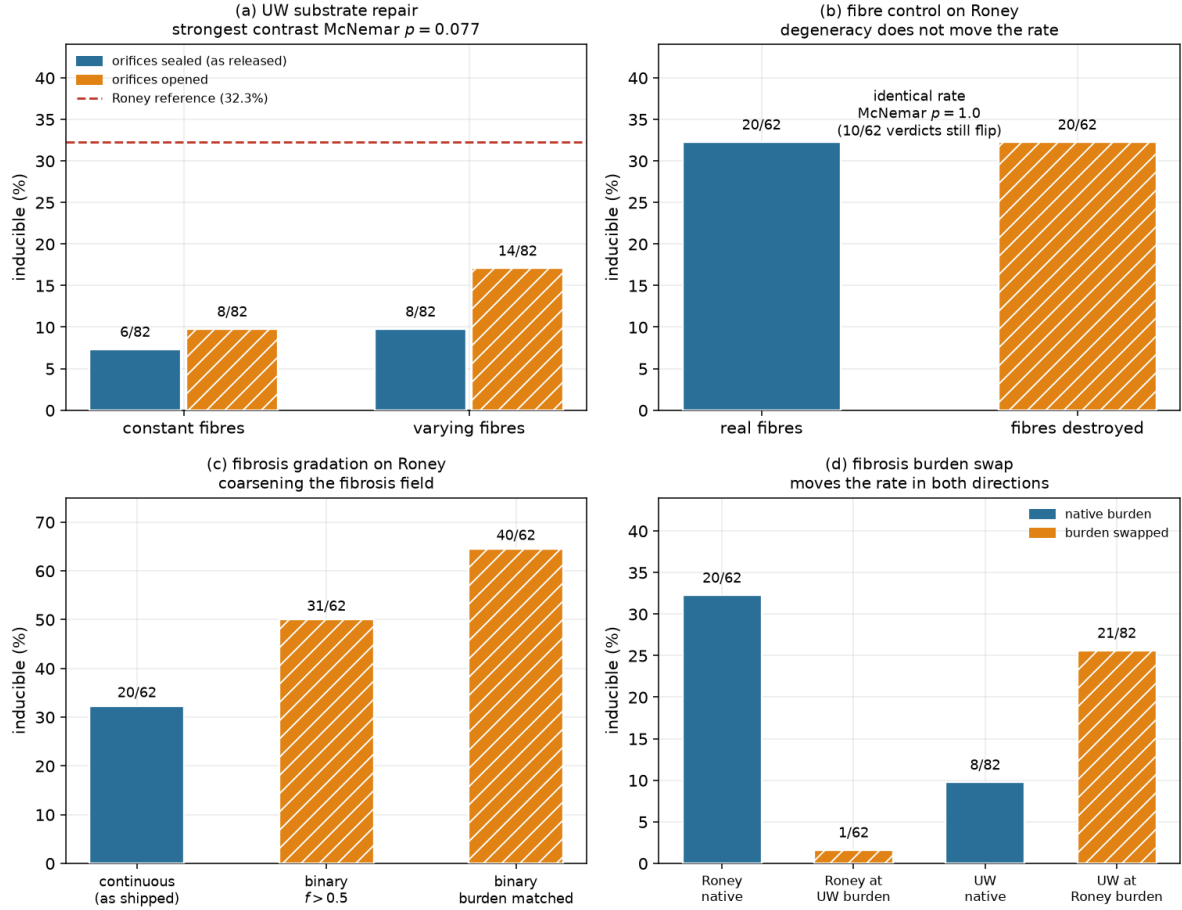


Figure 2: Four controls on the substrate. (a) Repairing both UW defects raises inducibility but not significantly, and not to the reference level. (b) Destroying the fibre field on a cohort that has one leaves the rate exactly unchanged. (c) Destroying the fibrosis gradation while holding mean burden fixed roughly doubles the rate, the opposite of the predicted direction. Panels (a)–(c) therefore leave all three substrate hypotheses unsupported, which is weaker than excluding them, and none of the three accounts for the anomaly. (d) Swapping fibrosis *burden* between the cohorts moves the rate in both directions and resolves the anomaly: Roney at UW’s burden collapses to 1/62, UW at Roney’s rises to 21/82. The dashed line is the Roney reference, 32.3%.

Taken together the three substrate controls leave the anomaly unexplained, and they tighten rather than loosen it: of the two defects we can manipulate, one does not move the rate at all and the other, imposed deliberately, moves it upward. None of the three establishes an effect, which is weaker than excluding one, and none of them points at a substrate *defect* as the cause. That is what redirected the search: from what might be broken in these meshes to what they contain.

Control 4: the cohorts do not carry comparable fibrosis. Having exhausted the defects, we measured what the solver is actually given. Coarsened to the same 2000 nodes, mean fibrosis is 0.349 on Roney and 0.244 on UW, and the fraction of tissue above half-fibrotic is 0.310 against 0.239. The UW cohort carries roughly 30% less fibrosis. The frozen protocol’s own calibration reports a rank correlation of 0.75 between fibrosis and sustained reentry, so a cohort with less of it should be less inducible. That is not an anomaly; it is the labeller behaving as specified.

Swapping the burden between cohorts confirms it in both directions (Fig. 2d). Rescaling Roney’s fibrosis multiplicatively to UW’s level, which preserves the spatial pattern and the

gradation and moves only the level, collapses inducibility from 20/62 to 1/62 (Fisher exact $p = 4.0 \times 10^{-6}$). Rescaling UW’s upward toward Roney’s raises it from 8/82 to 21/82 ($p = 0.013$). Both control arms reproduce their native rates exactly. The effect is large, symmetric, and an order of magnitude better supported than any of the three substrate hypotheses.

One detail is worth stating because it bears on the original question. At matched burden the point estimates invert: Roney rescaled to 0.241 gives 1/62 while UW at 0.248 gives 8/82. On one and eight events that contrast does not establish a reversal and we do not claim one; what it does show is that the UW cohort is not intrinsically resistant to reentry. Most of the apparent anomaly is accounted for by a difference in how much fibrosis the two released datasets contain, and the two released defects, real as they are, were never the explanation; what the burden account does not cover is openCARP’s UW rate, which the same rescaling leaves at 0/30 (Section 4.5). We had spent three controls looking for something broken before checking whether the two cohorts were comparable in the first place, which is the more general lesson: an out-of-range aggregate is a hypothesis about the pipeline, but it is a hypothesis about the inputs first.

4.5 An independent solver as a robustness check

Sources, all under `results/`: `opencarp_full_cohort.json`, with the per-cohort and chance-corrected concordance in `opencarp_concordance_detail.json`; all three calibration attempts, including the two rejected, are in `opencarp_calibration_history.json`, with the discretisation and propagation checks in `opencarp_bin_sensitivity.json` and `opencarp_uw_propagation.json`.

Every in-silico result above rests on one phenomenological labeller, a monodomain Mitchell–Schaeffer solver substituted for the pre-registered openCARP ground truth when openCARP could not be installed. That substitution is now testable: openCARP runs, and rather than replace the labeller with it — which would invalidate every number at once — we run both and report their agreement.

Calibration. openCARP ships the same Mitchell–Schaeffer membrane model, so the comparison is between solvers rather than between physics, but its defaults are not ours. Two quantities were matched to the frozen configuration, each measured independently of the endpoint. Action-potential duration: openCARP’s default `tau_close` 150 gives $APD_{90} = 277$ ms, against the frozen 218 ms healthy and 121 ms fibrotic; measured on this build, `tau_close` 110 $\rightarrow$ 209 ms and 55 $\rightarrow$ 113 ms, so $115 \times (1 - 0.5f)$ reproduces both within about 2%. Conduction velocity: the shipped conductivity gives 0.344 m/s against the frozen band of 0.4–1.2 m/s, which is a wavelength of roughly 75 mm in an atrium of ~ 110 mm characteristic length and makes reentry far too easy. Raising the intracellular conductivity alone saturates, because the monodomain conductivity is the harmonic mean of the intra- and extracellular values; scaling both gives 0.852 m/s at $g_{il} = 1.00$, and we adopt 1.05.

Two configurations were rejected before this one, at 90% and 76.7% inducible with *negative* fibrosis–verdict correlations, the signature of rate-dependent block rather than reentry. We record that because it is the protocol working: no endpoint parameter was adjusted at any stage, so the calibration gate remained able to fail, and it did.

Result. On the calibrated configuration the pre-registered gate is met on Roney: 22/100 inducible (22.0%, inside the 10–40% band) with a fibrosis–verdict rank correlation of +0.404, against the frozen solver’s 0.75. It is not met on UW, where openCARP returns 4/82 inducible (4.9%, below the band) with a rank correlation of only +0.079. Across all 182 subjects the two solvers agree on 156, or 85.7%. That figure is raw rather than chance-corrected, and on an outcome this imbalanced the correction matters: Cohen’s κ is 0.536, and the two solvers

agree on only 21 positives. Splitting it by cohort shows why the pooled number should not be read as a tissue-level result. On Roney, where the gate is met, raw agreement is 82/100 (82.0%) at $\kappa = 0.561$; on UW, where it is not, raw agreement is *higher* at 74/82 (90.2%) but κ falls to 0.287, because at a 4.9% and 9.8% event rate almost all of that agreement is shared negatives. The pooled figure is therefore a statement about the protocol across both cohorts rather than about the tissue, and the arm that looks most concordant is the one carrying the least information. The asymmetry is the substantive part. Of the 26 subjects the two solvers disagree on, openCARP calls only 5 inducible where the monodomain solver does not, while the reverse happens 21 times; over the whole cohort openCARP labels 26/182 against monodomain’s 42/182. That imbalance is systematic rather than sampling noise (McNemar exact $p = 0.0025$), so the two solvers differ in level and not only in which subjects they pick. The stand-in is therefore not under-calling relative to an independent reaction–diffusion solver — it is the more liberal of the two, which makes the incremental comparisons reported above conservative rather than inflated. (On the 60-subject calibration subset the discordance was entirely one-way, 0 against 5; that does not hold on the full cohort and the figure quoted here is the full-cohort one.)

What this check does not establish. Two limits are worth stating. The fibrosis field must be discretised into conductivity classes for openCARP, and the verdict is unstable at five classes (1/20) though identical at ten, twenty and forty (7/20, with complete per-subject agreement), so the operating choice of twenty sits inside the converged regime but the coarse limit is not benign. More importantly, openCARP returns 4/82 on the UW cohort, and unlike the monodomain solver this is *not* recovered by raising that cohort’s fibrosis burden: rescaling UW upward leaves the rate at 0/30. Two mundane explanations are excluded. It is not a propagation failure — on UW meshes the wave reaches 99.4% of nodes and peak depolarisation averages 0.83, so activation spreads across the surface and then dies, rather than never leaving the stimulus site. Nor is it the coarser discretisation: UW meshes carry a mean edge of 3.18 mm against Roney’s 2.64 mm at the same node budget, but the convergence study shows coarser meshes *over-call* (58.3% at 1500 nodes), so that difference predicts a higher UW rate, not a near-zero one.

What remains is a genuine substrate difference that the burden rescaling does not capture, and we do not know what it is. The monodomain burden result (Section 4.4) is established in both directions on that solver and is unaffected by this; what is not established is that openCARP’s near-zero UW rate has the same cause. We report the discrepancy rather than resolve it, and restrict the concordance claim to the cohort where both solvers are calibrated. It is worth noting which way this cuts: a robustness check that agreed everywhere would be less informative than one that disagrees somewhere and says so.

4.6 The labeller is reproducible in aggregate and unstable per subject

The controls expose something about the ground truth that matters more than the substrate defects do. Destroying the Roney fibre field holds the aggregate rate at exactly 20/62 while flipping 10/62 individual verdicts. Repairing the UW substrate retains only two of its six original positives. The six-tier convergence study reported below finds only 7/24 subjects consistent across resolutions. The aggregate inducibility rate is therefore a stable quantity, and the per-subject verdict is not.

This is a statement about the endpoint, not an aside. The pre-registered comparison is scored on per-subject labels, and non-differential noise in a binary outcome attenuates any incremental AUC toward zero. Part of the null we report is thus a property of the labeller rather than of the biomarker, and we cannot cleanly separate the two with the present ground truth. It also sets a floor on what any feature could achieve here, which is a limit on the whole class of studies that score spectral or structural features against a phenomenological inducibility

verdict, not on SFI alone. The pre-registered openCARP ground truth, substituted here for a monodomain Mitchell–Schaeffer solver, would need to be run before the residual is attributable.

4.7 The validity radius: why the single-vector estimator fails

The relative error of the first-order SFI scales as $\Theta(\rho)$ and crosses 10% at $\rho^* \approx 3$; real atria sit at median $\rho \approx 2422$. This proves the single-vector estimator is numerically invalid on real tissue and φ_2 is ill-conditioned, but (as stated in Section 2.4) it does not by itself prove non-predictiveness; the predictive null of Section 4.1 is a separate, label-dependent finding, and Section 4.2 shows redundancy accounts for only part of it.

4.8 The ρ scaling law

The gap collapses with system size by Weyl’s law [6]: for a d -dimensional domain the low Laplacian eigenvalues scale as $\lambda_k \sim (k/N)^{2/d}$, so ρ grows lawfully with resolution.

Table 2: Fitted gap-scaling exponents (gap vs. N at fixed mean degree, `results/rho_scaling.json`, five-point fits).

medium	fitted gap exponent	reference
real atrial surface (2-manifold)	$N^{-1.05}$	Weyl 2-manifold -1.0 (within 5%)
2-D random-geometric	$N^{-1.81}$	faster than naive $2/d$
3-D random-geometric	$N^{-0.84}$	faster than naive $2/d$ (non-monotone series)

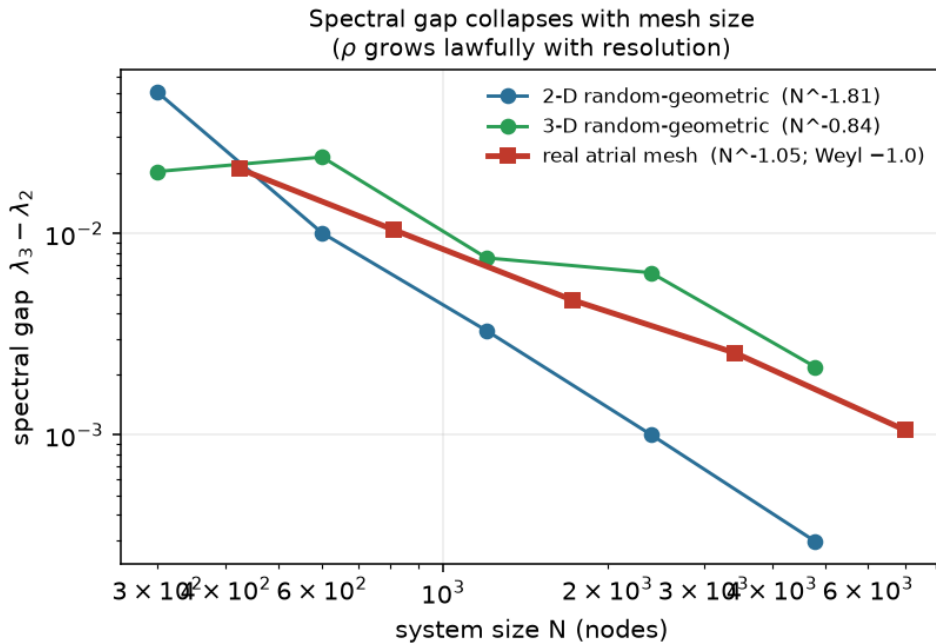


Figure 3: The gap collapses with mesh resolution, so the estimator degrades *more*, predictably, as the medium is meshed finer, the opposite of “more resolution helps.” Only the real atrial 2-manifold cleanly matches the Weyl exponent (-1.05 vs. -1.0); the random-geometric exponents are reported as “at least as fast as Weyl,” not law confirmations, and use the unweighted combinatorial Laplacian.

4.9 Localization transfers across three media (weakly)

The *strict* pre-registered localization endpoint is null for every field, including $|\nabla\varphi_2|$: its median argmax-geodesic error is 0.472 of the maximum geodesic extent (48.3 mm), with a bootstrap 95% CI of $[0.350, 0.656]$, and it is *not* below the spatial-null 5th percentile 0.406 (rotational/torus-shift permutation $p = 0.229$). The interval straddles that threshold, so on the 20 localized subjects the endpoint is not met and is also not resolved. What survives is weaker: $|\nabla\varphi_2|$ is a *rank* localizer (the true origin sits in its top $\sim 18\%$) that beats graph-centrality baselines (Perron, weighted degree, deleted by measured correlation) and holds across media.

Table 3: Fiedler-gradient origin-rank localization across dynamical media. *Origin rank* is the fraction of nodes whose localizer value is at least the value at the true reentry origin, so 0 means the origin is the global maximum of the field, 0.5 is chance, and 1 means it is the minimum; it is scale-free and comparable across media of different size. The companion metric used on real anatomy, *argmax-geodesic error*, is the shortest-path distance along the surface from the localizer’s argmax to the simulator’s reentry origin, expressed as a fraction of the mesh’s maximum geodesic extent rather than in millimetres (median 0.472, or 48.3 mm in absolute terms; `results/gm2_metrics.json`). Lower rank is better; a perfect localizer would place the origin at rank 0. The null-draw column counts resampled random-origin draws whose mean origin-rank is at or below the observed one, out of 300 draws per arm (each subject contributes up to 300 random nodes, and the pooled null is resampled to the smallest such sample). It is zero in every arm, which bounds the permutation p below $1/300 \approx 0.003$ but does not establish a magnitude: the resolution of the draw count, not the strength of the effect, is what the zero reports.

medium	$ \nabla\varphi_2 $ origin-rank	null draws $\leq$ obs.	fibrosis rank	n origins
FHN networks (100k)	0.176	0	~ 0.10	36,540
Kuramoto oscillators (500)	0.091	0	0.147	243
cardiac monodomain	~ 0.28	0	(co-localizes)	only ~ 20

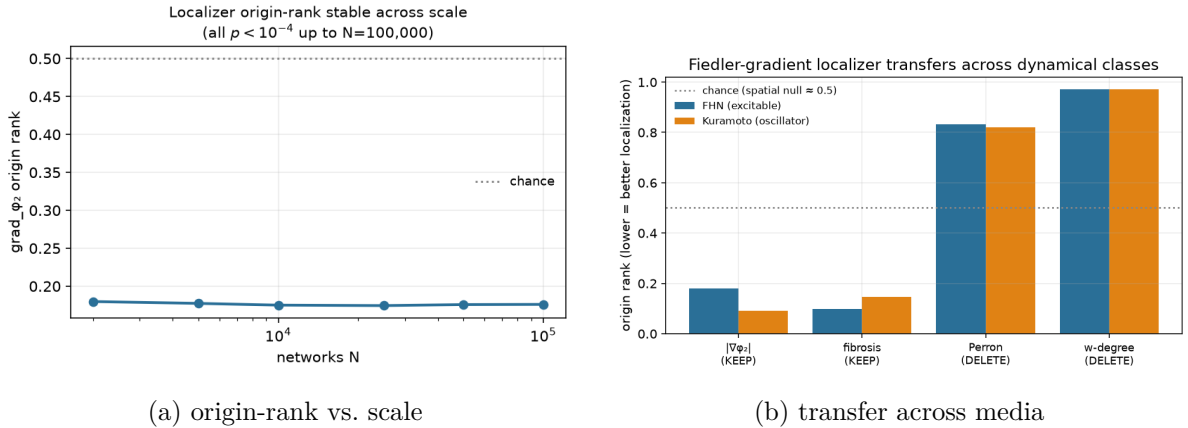


Figure 4: The weak-rank localization pattern replicates across excitable *and* oscillator dynamics. Three caveats apply: (i) the strict endpoint is null, so $|\nabla\varphi_2|$ is *not* a precise localizer; (ii) the substrate field co-localizes, and a controlled dissociation experiment shows why (in a diffusively-coupled medium the Fiedler bottleneck *is* the low-conduction lesion), so $|\nabla\varphi_2|$ matches substrate imaging rather than beating it, and substrate is the better localizer in the FHN arm; (iii) the cardiac arm rests on only ~ 20 origins.

4.10 The falsification protocol: what caught the false positive

Each guard, removed, manufactures a specific false positive (`results/falsification_protocol.json`). Shape-family leakage inflates AUC by +0.117 (naive random-CV 0.756 vs. grouped 0.640 on a controlled replicated cohort); the effect-size gate catches the p -value trap (subspace-SFI over the competitor set, paired DeLong $p \approx 2 \times 10^{-24}$ at $N = 55,000$, yet $\Delta\text{AUC} = +0.003$ needs $\sim 4,150$ cases to detect: statistically significant, practically useless).

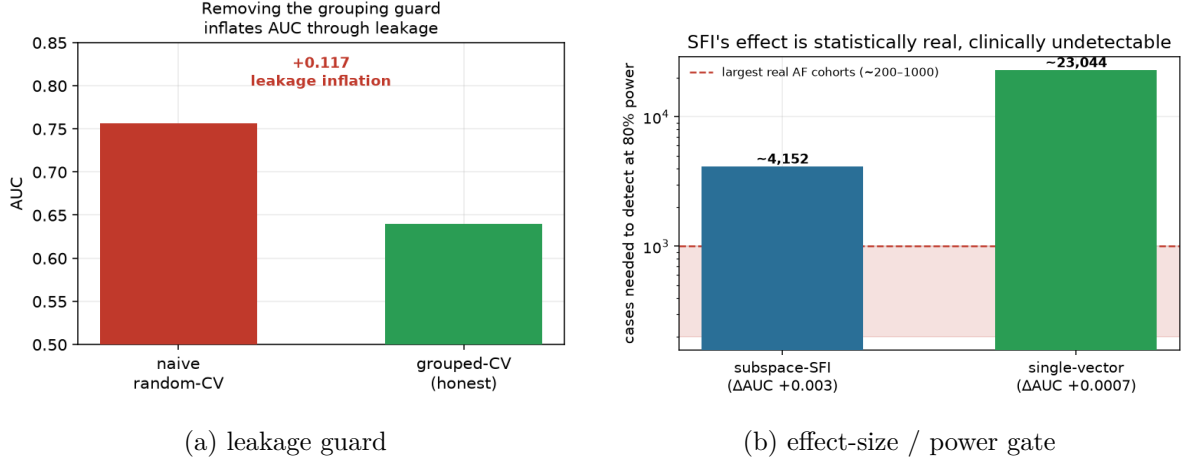


Figure 5: A naive analyst (random CV, $p < 0.05$, no effect-size gate, no spatial null) would have reported SFI as a working reentry biomarker. The pre-registered protocol correctly rejects it.

4.11 Uncertainty quantification

Morris elementary-effects screening (`results/e6_metrics.json`) finds the label most sensitive to diffusion d_0 ($\mu^* = 500.7$), then fibrosis density (147.8) and transverse conductivity (146.3). A Gaussian-process surrogate has leave-one-out $R^2 = 0.35$ (large residual anatomical/label variance beyond the screened knobs, a label-noise caveat, not a positive). Across uncoupling strengths $\Delta w \in \{0.18, 0.36, 0.54\}$ the grouped GBT increment is $\Delta\text{AUC} = 0.033\text{--}0.036$ with all $p \geq 0.26$. This sweep runs on *real* anatomy, not the synthetic media — it is the GM1 cohort, and on the original 62-mesh Roney subset that predates the extension to all 100 — so it speaks to the real-cohort null and not to the synthetic one. Every arm sits below the pre-registered 0.05 gate, but non-significance on its own does not make these arms null, and we report no interval for them, so they are indeterminate rather than a demonstration that the null is robust to Δw .

4.12 External inducibility-rate anchor (plausibility, not validation)

Clinically observed AF inducibility spans $\sim 5\%$ to $\sim 85\%$ depending on population, protocol aggressiveness, and the sustained-AF definition (Kumar et al. 2012 [14] report burst-sustained 14.8% vs. decremental-sustained 41.2% *in the same patients*). Our labeller’s Roney 34.0% (34/100) sits in the moderate-burst band (Marquardt et al. 2018 [15] report 30.6% in patients without prior AF), between the no-AF-control floor (5%, Oral et al. 2008 [16]) and the AF/aggressive ceiling (51%, Kawai et al. 2019 [17]), consistent with $\sim 42\%$ post-ablation (Liu et al. 2020 [18]). The UW 9.8% (8/82 on the cap-corrected substrate that every predictive result uses) sits just above the no-AF-control floor and far below the moderate-burst band, for a cohort that *is* AF patients. We flag it as anomalous. Section 4.4 traces it to a difference in fibrosis burden between the two releases rather than to biology or to the defects in the released UW substrate. Two substrate defects are documented in the Limitations, and repairing both

moves the rate from 6/82 to 14/82, a difference that does not reach paired significance (McNemar exact $p = 0.077$, $n = 82$) and that retains only two of the six original positives; a direct control shows the fibre defect is not responsible for any of it. This is a base-rate *plausibility check*, *not clinical validation*.

4.13 Resolution convergence: the labeller does not converge in the tested window

Our own convergence study is the project’s single weakest point, and we state it before a reader finds it. On a completed six-tier study (Roney, $n = 24$ at each of 1500/3000/6000/12000/24000/48000 nodes; 144 simulations, `results/convergence_summary_roney.json`), the inducibility rate crashes from the coarse mesh (58.3% at 1500) to 16.7% at 3000, then appears non-monotone across the fine tiers (25.0%, 20.8%, 25.0%) and rises again to 37.5% at the finest affordable tier ($\sim 50k$ nodes), a 17-percentage-point spread over the last three tiers, so it has *not* settled.

How much of that pattern is real deserves stating, because $n = 24$ per tier buys little precision. Exact binomial intervals are [36.6, 77.9], [4.7, 37.4], [9.8, 46.7], [7.1, 42.2], [9.8, 46.7] and [18.8, 59.4] percent across the six tiers. These are marginal intervals: the same 24 subjects are simulated at every tier, so each describes one tier’s own precision and discards the pairing, and overlap between two of them is not a test of their difference. We use them only for the negative claim. Every tier from 3000 nodes upward overlaps every other, so the apparent wiggle among the fine tiers is *not* distinguishable from sampling noise and we do not claim it is a real non-monotonicity.

The paired comparison the intervals cannot make is available, because the same 24 subjects run at every tier: for each adjacent pair the 2×2 table is fixed by the two marginals and the agreement count, so an exact McNemar test follows without any further simulation (`results/convergence_paired_tests.json`). It separates the two halves of the pattern cleanly. The coarse-mesh drop is real: 11 subjects lose their verdict between 1500 and 3000 nodes against 1 that gains one, $p = 0.0063$, and it survives Holm correction across the five adjacent comparisons ($p = 0.032$). None of the four fine-tier transitions is resolvable ($p = 0.375$ to 1.000 unadjusted, all 1.000 after correction). What the study establishes is therefore sharper than the intervals alone allow, and still sufficient for our purpose: the coarsest tier demonstrably over-calls, while across the fine tiers the rate cannot be shown to have converged and no tier can be declared the settled value. A study that could distinguish these tiers would need several hundred subjects per tier, which is roughly two orders of magnitude more simulation than the 144 runs here.

Table 4: Inducibility rate vs. mesh resolution (Roney, $n = 24$ per tier). The rate does not converge in the tested window; adjacent-tier per-subject verdict agreement is itself non-monotone (50%, 75%, 71%, 63%, 79%) and the rate keeps drifting. At $n = 24$ per tier the exact binomial intervals overlap for every tier from 3000 nodes upward, so the ordering among the fine tiers is not resolvable; the claim supported is non-convergence, not a specific shape.

nodes	1,500	3,000	6,000	12,000	24,000	48,000
inducibility rate	58.3%	16.7%	25.0%	20.8%	25.0%	37.5%
adj.-tier agreement	—	50%	75%	71%	63%	79%

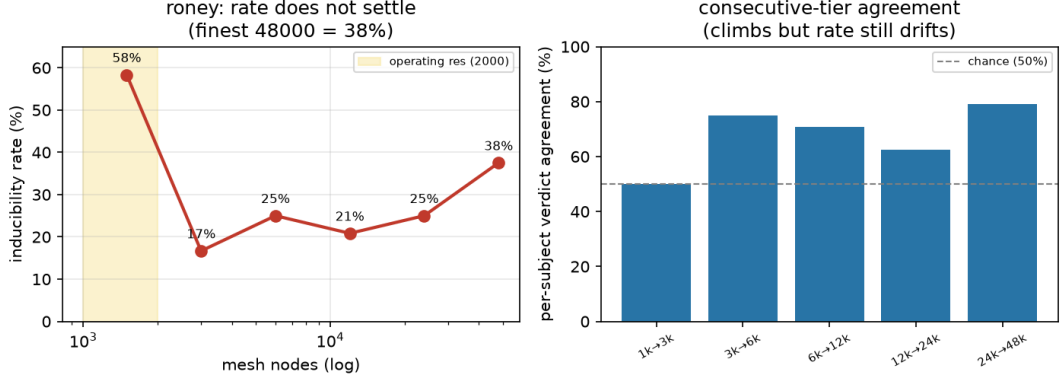


Figure 6: Left: the inducibility rate does not settle; the finest tier (48,000 nodes = 37.5%) sits above the mid tiers. Right: adjacent-tier per-subject verdict agreement is non-monotone (50%, 75%, 71%, 63%, 79%) while the aggregate rate still drifts. Only 7/24 subjects are fully consistent across all six tiers.

The convergence finding limits the absolute numbers but not the comparative claims. The rate does *not* collapse to zero; the phenomenon is real; it is the *magnitude* that is resolution-sensitive up to $\sim 50k$ nodes. Because the main results ran at 2,000 nodes, every *absolute* number (inducibility rate, competitor AUC, localizer origins) is coarse-mesh-provisional pending the deferred openCARP validation. What this does *not* touch is what we lead with: (a) the *relative* SFI-vs-competitor comparison, since both arms score the *same* labels, so the shift in *absolute* AUC is common to them; and (b) the label-free estimator-validity argument (ρ is computed from the graph Laplacian, no labels at all). Sharing labels does not make the relative comparison unbiased: per-subject label noise attenuates the incremental ΔAUC toward zero (Section 4.6), so that comparison is conservative.

5 Discussion

This work makes four contributions. (1) A pre-registered, leakage-controlled, effect-size-gated evaluation of a plausible spectral biomarker: a clean null at synthetic scale (10^5 networks), where the mechanism remains inferred, and a measured null on 182 patients of real anatomy, where it is measured rather than inferred (Section 4.2, on those same 182 subjects: the aggregate SFI summaries are reconstructible from the competitors at $R^2 = 0.970$, the extreme and edge-level columns are not, and their unique variance is unpredictable). (2) A label-free validity criterion ρ for the single-vector estimator: it proves φ_2 is ill-conditioned on real tissue ($\rho \approx 2422 \gg \rho^* \approx 3$), a portable a priori check for any Fiedler-based marker, with the caveat that ρ bounds estimator accuracy, not predictive content. (3) A weak-but-real, purely connectivity-derived rank localizer that beats centrality baselines across three media but does not beat substrate localization, which is the better localizer in the FHN arm (Table 3); the strict pre-registered localization endpoint is null. (4) A demonstration that standard methodology manufactures the exact false positives this protocol catches (leakage +0.117 AUC; the p -value trap).

5.1 Four corrections, and what caught each

Four claims in earlier versions of this work did not survive, and because they are referred to individually where they arise, we set them out together here.

The first was a positive result. A pre-specified additional feature-set comparison reported that SFI beat a fibrosis-heterogeneity baseline on the combined real cohort ($\Delta AUC = +0.103$, $p = 0.009$), which we read as showing SFI is not merely re-encoded fibrosis. Extending the

Roney arm from 62 to all 100 released meshes reduced it to $+0.017$ ($p = 0.452$). What caught it was more data, and what made the original vulnerable was that it rested largely on one cohort’s $+0.340$ over a near-chance baseline on six events.

The second was a causal attribution. We reported that the degenerate fibre field in the UW release was the likely reason that cohort is so rarely inducible. What caught it was a direct control: destroying the fibre field on a cohort that has an intact one left the inducibility rate at exactly $20/62$, $p = 1.0$. The explanation was plausible, mechanistically motivated, and wrong, and no amount of further argument would have revealed that.

The third was a set of geometric figures. We stated that element tag 164 formed exactly five components, had exactly zero adjacency to fibrosis, and left Euler characteristic -3 on removal. All three were generalised from a single mesh or from a rounded mean. What caught them was writing regression tests over the whole cohort, two of which failed on first run. The identification of tag 164 itself survived and is better supported by the census than by the original figures.

The fourth was the primary endpoint itself. An earlier version reported a combined real-cohort ΔAUC of $+0.051$ and read the endpoint as missed only on significance. Correcting the UW substrate at the original 144 subjects brought that to $+0.036$; extending the Roney arm from 62 to all 100 released meshes brought it to $+0.012$, so the larger share of the drop was sample size rather than substrate. What caught it was the substrate audit followed by the cohort extension (Section 4.1).

We note the pattern rather than each instance. Every one was caught by a control, a larger sample, or a test that this project designed and ran against its own results before submission; none was found by a reader. The first correction is also a worked example of what the paper argues elsewhere. Naive methodology — a $p < 0.05$ threshold with no effect-size gate, no multiplicity accounting and no replication on more data — would have reported $\Delta\text{AUC} = +0.103$ at $p = 0.009$ as a finding. It was produced by our own pipeline and it was false, and the protocol that rejected it is the same one applied to the primary endpoint throughout.

The validity screen is domain-general. Spectral-graph fragility markers are used for seizure-focus localization, brain connectomics, and power-grid vulnerability. The same precondition applies: compute $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$; if $\rho \gg 1$, the first-order spectral marker is outside its validity radius, so its *numerical* $\Delta\lambda_2$ estimates and the stability of φ_2 cannot be trusted. That is a statement about the estimator, not about whether the feature ranks cases usefully, which must still be tested against competitors.

We are equally explicit about what we do not claim. The mathematics is classical; no new theorem. ρ does *not* prove SFI is non-predictive, only that the single-vector estimator is ill-conditioned; the predictive null is separate from it, and is only partly redundancy (Section 4.2). The real-cohort result *is* a measured null in the sense that the interval excludes the pre-registered gate, but it does *not* exclude a small effect: the lower bound is -0.017 , so an increment of a few thousandths remains compatible with the data and would need a far larger cohort to resolve. The localizer is *not* precise (strict endpoint null) and does *not* beat substrate imaging. The ρ scaling “law” is a clean match to Weyl only on the real atrial 2-manifold. Labels are a monodomain simulator, not clinical outcomes and not openCARP.

6 Limitations

1. The labeller is strongly resolution-sensitive and does not converge in the tested window (Section 4.13): the rate drops from 58% at 1500 nodes to 17% at 3000 and rises to 37.5% at $\sim 50\text{k}$ nodes, above the 34.0% rate at the 2,000-node operating resolution, so the direction of the residual bias there is not established. At $n = 24$ per tier the intervals overlap for every tier from 3000 nodes upward, so no ordering among the fine tiers is resolvable and we do not claim the wiggle there is real; only $7/24$ subjects are fully consistent across all six tiers. All *absolute* rates and AUCs are coarse-mesh-provisional; the *relative* comparison scores the

same labels in both arms, so the shift is common to them, though per-subject label noise still attenuates the incremental ΔAUC toward zero.

2. Simulator, not clinical, labels: the pre-registered openCARP ground truth was substituted with monodomain Mitchell–Schaeffer, a logged deviation.
3. A clinical endpoint family *is* pre-registered and is deliberately not reported here. Per-patient two-year post-ablation recurrence outcomes for the 82 UW patients were shared with us after the simulator analysis was complete, and §8 of the pre-registration was frozen and tagged before any feature was placed alongside the outcome column; at the time of writing no such association has been computed. Two things follow that we state now rather than after seeing an answer. The outcome column is restricted-use human-subject data, not covered by the CC0 that applies to the meshes, so it is not redistributed with this work and only aggregate statistics derived from it may be published. And the study is underpowered for that endpoint by construction: on the observed 48/82 event rate, power at the $\Delta\text{AUC} \geq 0.05$ gate used for the simulator endpoint is 0.24 at best and 0.13 under a weaker shared-noise assumption, so that gate is *not* reused there. The minimum detectable effect is $\Delta\text{AUC} \approx 0.10$ to 0.15, and 80% power at 0.05 would need 371 to 859 patients (`results/power_real_outcomes.json`). A null below that band will be reported as inconclusive, never as evidence that SFI adds nothing clinically.
4. The UW cohort’s released substrate carries two defects, detailed with their controls in Section 4.4: a fibre field that is a constant vector in all 164 meshes, and element tag 164, which is the caps over the atrial orifices and which we had been treating as half-fibrotic tissue. Repairing both raises inducibility only from 6/82 to 14/82, and none of the three substrate explanations for the cohort’s low rate is supported by direct control; Control 4 attributes the low rate instead to a fibrosis burden roughly 30% lower (0.244 against Roney’s 0.349), reproduced by swapping the burden in either direction. Its regions additionally rest on a PCA surrogate rather than registered atrial coordinates. The cohort’s predictive test is uninformative about SFI, and what remains open is narrower: openCARP returns 4/82 here and the same rescaling does not recover it (0/30, Section 4.5).
5. The inducibility label is stable in aggregate but noisy per subject, which attenuates every ΔAUC we report. Destroying the Roney fibre field holds the rate at 20/62 yet flips 10/62 individual verdicts; repairing the UW substrate retains only 2 of its 6 original positives. That compounds the mesh-convergence instability above, where just 7/24 subjects were consistent across six resolutions. Non-differential label noise biases incremental AUC toward zero, so the null we report is partly a statement about the labeller and not only about SFI.
6. ~ 30 DeLong comparisons were run on the real cohorts. The pre-registration provides no multiplicity plan for that family, so every p -value we report is unadjusted; on the full 182-subject corrected analysis none reaches the pre-registered $\alpha = 0.05$. The smallest is $p = 0.121$ (`fibrosis_vs_+SFI`, gradient boosting, UW), and that one sits on a base AUC of 0.562 with eight positives, so it is uninformative rather than borderline. An earlier version reported this family’s combined estimate as the study’s one positive at $p = 0.009$; it is withdrawn (Section 5.1).
7. ρ^* and the random-geometric gap exponents are empirical; only the real atrial 2-manifold cleanly matches the Weyl exponent.

7 Conclusion

A closed-form spectral fragility index does not add practically meaningful reentry-inducibility prediction beyond existing features, either at synthetic scale or on 182 patients of real anatomy,

where the combined estimate is $+0.012$ with an interval that excludes the pre-registered gate. The mechanism is decomposable and only half of it is redundancy: the aggregate SFI summaries are reconstructible from the competitor set at $R^2 = 0.970$, while the extreme and edge-level columns carry variance of their own that simply does not predict the label. Independently of the predictive question, the single-vector estimator is ill-conditioned on real tissue ($\rho \gg 1$), and lawfully more so as meshes are refined.

The core result is not a discovery but a carefully-scoped boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how readily standard methodology would have reported a working biomarker — a false positive this project produced from its own pipeline, at $\Delta\text{AUC} = +0.103$ and $p = 0.009$, and then withdrew. What the study cannot settle is the absolute inducibility rate, which does not converge within the mesh resolutions we could afford; the comparative claims, which score identical labels in both arms, do not depend on it. An independent solver run alongside the labeller agrees with it on 85.7% of subjects and is the more conservative of the two, which bounds how much of the null could be an artefact of the stand-in.

A Formal (proof-assistant) verification of the SFI algebraic core

The algebraic core of the SFI edge-sensitivity identity is re-proved inside the Coq 8.18 proof assistant and checked by its small trusted kernel, an assurance level above a computer-algebra system. The primary file (`formal/sfi_edge_identity_Q.v`) works over the exact rationals $\mathbb{Q}$, so `Print Assumptions` reports “*Closed under the global context*” for every theorem: zero axioms. A mathematically identical companion over the reals $\mathbb{R}$ (`formal/sfi_edge_identity.v`) inherits Coq’s two standard real-number-library axioms and is kept only to show the statements hold over $\mathbb{R}$.

Table 5: Kernel-checked theorems ($v = e_i - e_j$ the edge vector, $E_{ij} = \partial L / \partial w_{ij} = vv^\top$). T1–T5 are axiom-free over $\mathbb{Q}$.

name	statement	meaning
T1_rank_one_quadratic_form dot_edge (T2)	$\varphi^\top (vv^\top) \varphi = (v \cdot \varphi)^2$ $(e_i - e_j) \cdot \varphi = \varphi_i - \varphi_j$	quadratic form of a rank-one edge matrix the edge vector selects the potential difference
T3_sfi_edge_sensitivity	$\varphi^\top E_{ij} \varphi = (\varphi_i - \varphi_j)^2$	the SFI edge term
T4_sfi_term_nonneg	$0 \leq w \Rightarrow 0 \leq w(\varphi_i - \varphi_j)^2$	each SFI contribution is non-negative
T5_aggregate_nonneg	$\sum(\text{nonneg terms}) \geq 0$	SFI is a Dirichlet energy; the Laplacian form is PSD

Combined with the classical Hellmann–Feynman theorem $\partial \lambda_2 / \partial w_{ij} = \varphi_2^\top (\partial L / \partial w_{ij}) \varphi_2$ (Ghosh–Boyd 2006 [2], an analysis result we cite, not re-prove), T3 is exactly the closed form $\partial \lambda_2 / \partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$. Lean/Mathlib was attempted; its toolchain binaries are distributed only as GitHub release assets, which the compute environment’s egress policy blocks with a hard 403, so Coq provides the equivalent kernel-checked guarantee. Reproduce with `coqc formal/sfi_edge_identity_Q.v`.

B Reproducibility

All stochastic steps are seeded; the exact configuration and commit hash accompany every reported number. Results in this paper were produced on Python 3.12.10 with NumPy 2.5.1, SciPy 1.18.0, scikit-learn 1.9.0, pandas 3.0.5 and Matplotlib 3.11.1; the openCARP cross-check used openCARP v19.0. The per-subject record cache is keyed by a hash of the frozen

configuration, so changing any frozen constant invalidates it rather than silently reusing stale labels. Numeric verification: `python scripts/verify_sfi_identity.py` and `python scripts/verify_math_rigor.py` (sympy symbolic, mpmath 40-digit, exact-rational, interval-certified, and cross-solver checks). Convergence: `python scripts/convergence_analyze.py roney`. Figures are regenerated by `scripts/make_paper_figures.py`. Datasets: Roney LA cohort (Zenodo 5801337) and UW/Boyle meshes (Dryad 10.5061/dryad.kkwh70sg0); both are public and de-identified.

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